Phase Reconstruction in a GAN-less Neural Audio Codec: A Failure-Mode Analysis and Rate-Distortion Study

Ruiting Ma
Master of AI, Monash University
rmaa0045@student.monash.edu

Code: https://github.com/RuitingMa/mini-codec

Abstract

We implement an Encodec/SoundStream-style neural audio codec from scratch and use it to characterise the failure modes of *GAN-less* training at low bitrate. A 5.4M-parameter encoder, residual vector quantizer (RVQ), and decoder are trained on `LibriSpeech train-clean-100` with a time-domain ℓ_1 plus multi-scale mel-spectrogram loss; no adversarial discriminator is used. On `test-clean` (256 utterances) at 3.2 kbps, per-sample multi-scale mel-ℓ_1 is tightly clustered (std ≈ 0.009) while per-sample SI-SDR varies over a 30 dB range, identifying the failure mode as phase / fine-time-structure error rather than spectral envelope error. A pre-registered ablation tests whether HuBERT-base layer-6 features can substitute for adversarial training in supplying phase-aware gradient: the additive variant is statistically indistinguishable from the baseline on SI-SDR (median delta -0.82 dB at $\sim$1 dB SE), while a swap-STFT control collapses to median -30 dB, together ruling out HuBERT-style ASR-pretrained features as a phase-recovery signal because they encode envelope but not phase. A rate-distortion sweep at four bitrates (1.6/3.2/6.4/12.8 kbps, all other parameters held fixed) yields a curve linear in log-bitrate at $\sim$6 dB SI-SDR per octave, with the gap to Encodec's published with-GAN numbers narrowing monotonically from $\sim$21 dB at 1.6 kbps to $\sim$13 dB at 12.8 kbps, suggesting the marginal contribution of adversarial training is largest where the information bottleneck is tightest.

1 Introduction

Neural audio codecs based on encoder–quantizer–decoder pipelines (Défossez et al., 2022; Zeghidour et al., 2021; Kumar et al., 2023) have closed much of the perceptual quality gap to traditional codecs at low bitrate. Their reported quality numbers depend, however, on a multi-component loss that combines time-domain reconstruction, multi-scale spectrogram losses, and an *adversarial* discriminator term in roughly equal weighting (Défossez et al., 2022). Removing the adversarial component is known empirically to lose roughly 6 dB of SI-SDR (Défossez et al., 2022, §3.5), but the mechanism behind that gap, and whether non-adversarial substitutes could close it, is not well-characterised.

We therefore ask what a neural audio codec actually fails at when trained without adversarial loss, and what class of substitute losses could in principle replace adversarial training. Answering these questions requires a setup that isolates the encoder, quantizer, and reconstruction-loss design from the GAN dynamics, then probes those choices systematically. We build that setup, a 5.4M-parameter Encodec-style codec trained on `LibriSpeech train-clean-100` with a deliberately GAN-free scope, and use it to make three contributions.

The first is diagnostic. At the project's default 3.2 kbps operating point, per-sample multi-scale Mel L1 across `test-clean` is tightly clustered (std ≈ 0.009) while per-sample SI-SDR ranges over 30 dB, identifying the residual error after non-adversarial training as living in phase / fine-time-structure rather than in spectral envelope (§3). The second is a pre-registered ablation that asks whether features from a frozen self-supervised audio encoder (HuBERT-base layer 6) can supply the missing phase-aware gradient: an additive variant is statistically indistinguishable from the baseline on SI-SDR ($\Delta = -0.82$ dB at SE ~ 1 dB), and a swap-STFT anti-gaming control collapses to median -30 dB, ruling out a structural family of perceptual losses for reasons we develop in §4. The third is quantitative: a four-point bitrate sweep at $\{1.6, 3.2, 6.4, 12.8\}$ kbps yields a curve linear in log-bitrate at ~ 6 dB SI-SDR per octave, with the gap to Encodec's published with-GAN numbers narrowing monotonically from ~ 21 dB at the low-bitrate end to ~ 13 dB at the high-bitrate end, suggesting the marginal contribution of adversarial training is largest where the information bottleneck is tightest (§5).

2 Method

2.1 Architecture

The encoder is a 1D convolutional stack with weight-normalised (Salimans and Kingma, 2016) `Conv1d` layers and ELU activations. Downsampling factors $(2, 4, 5, 5)$ give a $200\times$ temporal compression, i.e. an 80 Hz frame rate at the project's 16 kHz native sample rate; channel count doubles at each down-sample; a final 1×7 projection emits a 128-dimensional latent. The decoder mirrors the encoder using `ConvTranspose1d`, with `output_padding=1` on odd strides so the round-trip preserves length to the sample. Compared to Encodec, the architecture trains on the 16 kHz `train-clean-100` subset (rather than 24 kHz mixed-domain data, so the encoder does not have to model music), uses no LSTM bottle-neck, and, as the central methodological choice for this work, uses no adversarial discriminator. Total trainable parameters: 5.4M, against Encodec's ∼14M.

2.2 Quantizer

We use a residual vector quantizer (RVQ; Zeghidour et al., 2021) with 4 layers and 1024 codes per layer at the default bitrate point. Codebooks are updated by EMA over assigned encoder outputs (no autograd through them) and seeded data-dependently from the first training batch. We add a *dual-trigger* dead-code restart that fires when either (i) a code has gone ≥ 20 consecutive steps without an assignment, or (ii) its EMA cluster size has decayed below 0.01. The streak-based trigger is necessary at our scale: with batch $\times$ segment frames $= 320$ and codebook size 1024, the steady-state EMA cluster size for a uniformly used code is only ∼0.31, and an EMA-only criterion at the canonical (Défossez et al., 2022) threshold of 2.0 would trigger continuously. Without these stability tricks, all four RVQ layers collapse to a single active code within ∼30 training steps.

2.3 Loss

The training objective for the baseline is a weighted sum of a time-domain ℓ_1 term, a multi-scale log-mel STFT term, and the RVQ commitment loss:

$$\mathcal{L} = 0.1 \cdot \|x - \hat{x}\|_1 + \mathcal{L}_{\text{STFT}} + \mathcal{L}_{\text{commit}}. \tag{1}$$

$\mathcal{L}_{\text{STFT}}$ is the multi-scale log-mel $\ell_1 + \ell_2$ distance over six STFT window sizes $\{64, 128, 256, 512, 1024, 2048\}$ (each scale with $\min(64, n_{\text{fft}}/8)$ mel bins, capped to keep the filterbank well-conditioned); and $\mathcal{L}_{\text{commit}}$ is

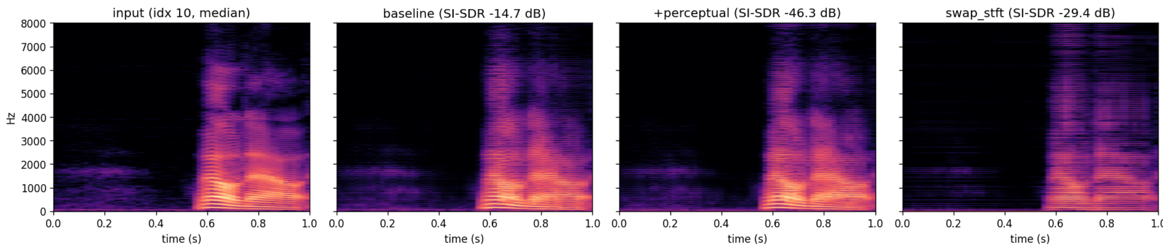


Figure 1: Spectrogram comparison of the input and reconstructions from baseline, additive perceptual, and swap-STFT variants on a median-SI-SDR utterance from `test-clean`. The envelope structure is preserved in baseline and `+perceptual` even when their SI-SDR values differ by 30 dB; `swap_stft` has visible horizontal banding consistent with phase artefacts. The median-rank utterance happened to be a `+perceptual` catastrophic outlier (SI-SDR -46.3 dB); its envelope nevertheless remains close to the input.

the standard RVQ commitment loss (van den Oord et al., 2017). The perceptual ablation in §4 adds an optional fourth term $\lambda \cdot \mathcal{L}_{\text{percep}}$. We use no adversarial loss anywhere in this work: our project question is what non-adversarial training can recover from this architecture, with a within-architecture GAN ablation reserved as future work.

3 Baseline failure mode

The baseline is trained for 50k steps at batch size 16 with Adam ($\eta = 3\times10^{-4}$, $\beta = (0.9, 0.99)$) on a single RTX 4090, taking roughly 35 minutes of wall time. Evaluation runs on the first 256 utterances of `test-clean` with deterministic crops and a fixed seed, scoring SI-SDR and the same multi-scale Mel-ℓ_1 used during training. At 3.2 kbps the baseline reaches SI-SDR median -14.10 dB (std 10.35 dB) and multi-scale Mel L1 median 0.309 (std 0.215). The standard deviation on SI-SDR is driven largely by silence-dominated outliers, where SI-SDR is sensitive to small reconstruction noise when the target is mostly zero, so the median is the more representative summary statistic.

The two metrics' *distributions* tell the more interesting story. Per-sample multi-scale Mel-ℓ_1 across the dumped subset clusters tightly at std ≈ 0.009, while per-sample SI-SDR spans 30 dB, three orders of magnitude apart in standard deviation. Spectral envelope is reproduced with high consistency across utterances; time-domain alignment is not. Inspection of the per-sample spectrograms (Fig. 1) confirms this directly. Formant structure, energy distribution, and onset locations of the input and the reconstruction match across the three variants we plot, but the per-sample waveforms drift in phase by several milliseconds even where the magnitude STFT is essentially identical. This "envelope reproduced, phase misaligned" signature is characteristic of GAN-less audio codecs at low bitrate.

4 Perceptual loss as a phase substitute? A pre-registered negative

If a frozen pretrained self-supervised audio encoder produces features whose stability requires phase information, then penalising the ℓ_1 distance between such features for x and $\hat{x}$ should supply the codec with the phase-aware gradient that time-domain L1 plus magnitude STFT does not provide. We test this hypothesis with HuBERT-base (Hsu et al., 2021) layer 6 features, chosen as the standard phonetic / acoustic mid-layer after a 5k-step layer sweep over $\{4, 6, 9\}$.

The experiment uses two pre-registered conditions, recorded in the project repository's experiment plan before training. The *additive* variant adds the perceptual term to the baseline loss with weight 1.0; the *swap-STFT* variant replaces the multi-scale mel term with the perceptual term while leaving everything

Table 1: HuBERT-base layer-6 perceptual loss vs. baseline, `test-clean` 256 utterances.

Variant	SI-SDR median (dB)	SI-SDR mean (dB)	Mel L1 median	Mel L1 mean
baseline	−14.10	−15.70	0.309	0.399
+perceptual (additive)	−14.92	−16.05	**0.301**	**0.396**
swap-STFT	−30.12	−32.04	0.693	0.864

else unchanged. The second variant is an anti-gaming control. If perceptual loss alone is too weak to support codec training, swap-STFT will collapse; if perceptual loss is independently strong, swap-STFT should match or exceed the baseline. Without this control, an additive variant that improves on baseline could equally well be reading off the STFT term that already does most of the work.

Results on the same 256-utterance `test-clean` pass appear in Table 1. The additive variant reaches median SI-SDR −14.92 dB, a 0.82 dB drop relative to baseline at a standard error of roughly 1 dB, well within the pre-registered "failed" threshold of a 1 dB lift in either direction. Mel L1 marginally improves (0.301 vs. 0.309), the only metric that moves systematically in the positive direction. The swap-STFT variant collapses to median SI-SDR −30.12 dB, a 16 dB regression from baseline at which the model is essentially producing noise.

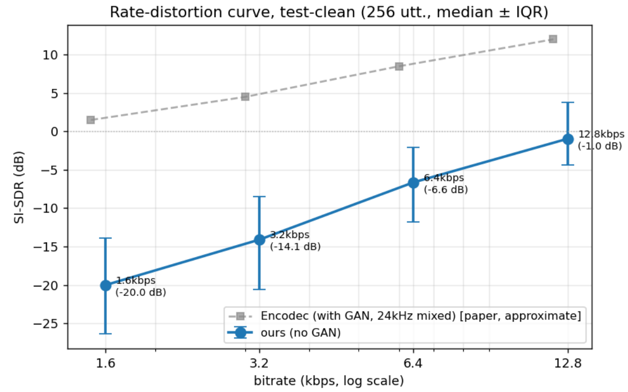
Read together, the two conditions rule out the most charitable interpretation of the additive result. If the perceptual loss were contributing useful phase signal that the STFT term simply absorbs, swap-STFT, where there is no STFT to absorb anything, should still train at least to baseline-level reconstruction quality. It does not. The perceptual loss is therefore not independently strong enough to replace STFT's spectral signal, and the additive variant's small Mel-ℓ_1 improvement (with no SI-SDR change) places HuBERT's contribution in exactly the spectral envelope dimension that multi-scale STFT already covers. We argue the cause is structural. HuBERT and similar SSL audio encoders are trained on masked-prediction objectives that can be satisfied using envelope / phonetic content alone; their representations need not encode phase, and our results suggest they in fact do not. A perceptual loss derived from such an encoder cannot supply phase-aware gradient regardless of layer choice, which closes off this entire family of substitutes for adversarial training.

5 Rate-distortion characterisation

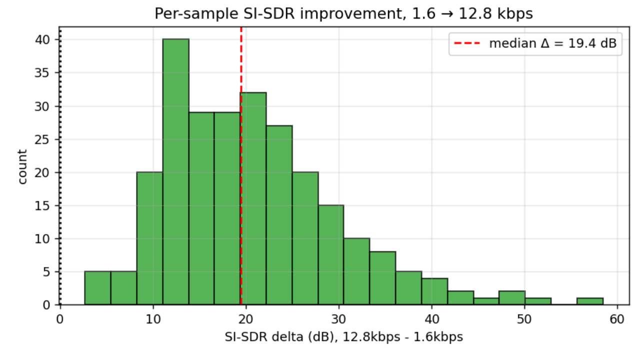
To characterise the codec's capacity at the chosen architecture, we sweep the number of RVQ layers $N \in \{2, 4, 8, 16\}$, holding everything else fixed. The resulting bitrates are $\{1.6, 3.2, 6.4, 12.8\}$ kbps. Each configuration is trained from scratch for 50k steps; evaluation is the same 256-utterance `test-clean` pass.

The resulting rate-distortion curve in Fig. 2a is approximately linear in log-bitrate, with per-octave SI-SDR gains of 5.93, 7.47, and 5.68 dB across the three intervals. Higher bitrates also produce more consistent reconstructions: the inter-quartile range narrows monotonically from 12.5 dB at 1.6 kbps to 8.2 dB at 12.8 kbps. Per-sample improvements are universal as well: Fig. 2b shows that the SI-SDR delta from 1.6 to 12.8 kbps is entirely positive across the 256 utterances (median +19.4 dB, range roughly +3 to +60 dB). No sample regresses with more bits, so the bitrate increase is being spent productively across the full distribution rather than fitting a few easy samples.

Comparison to Encodec's published numbers (Défossez et al., 2022, Table 4) is informative even though our setup is not a controlled ablation of theirs. The gap is $\{21.5, 18.6, 15.1, 13.0\}$ dB at the four bitrates, a monotonic narrowing of 8.5 dB across the sweep. Encodec's setup differs from ours in several respects



(a) R-D curve: SI-SDR median (±IQR) vs. bitrate. Encodec reference is approximate, from the paper's Table 4 (24 kHz mixed, with GAN); see text for caveats.



(b) Per-sample SI-SDR delta, 12.8 kbps minus 1.6 kbps. Every utterance improves; median +19.4 dB.

Figure 2: Rate-distortion behaviour of the codec at fixed architecture.

(adversarial loss, 24 kHz mixed-domain training data, a larger model, and a much larger training corpus), so we cannot read this trend as a clean measurement of the GAN contribution. Since adversarial training is, however, the most distinctive methodological difference between the two systems, the narrowing is consistent with the hypothesis that GAN's marginal contribution to perceptual quality is largest where the information bottleneck is tightest. Adversarial training is the mechanism best positioned to "fill in" detail the encoder physically cannot store; at 12.8 kbps, where the encoder has enough bits to encode that detail directly, the marginal value of the discriminator should plausibly be smaller.

An informal listening check by the author on a handful of 12.8 kbps reconstructions matched the metric picture: speech is fully intelligible and the spectral envelope is recognisable, while residual phase and timbre artefacts remain clearly audible. This is consistent with the codec sitting at SI-SDR ≈ 0 dB: good enough that the signal is recovered, not good enough that the reconstruction is transparent. A formal listening study is left to future work.

6 Discussion

Our codec at 12.8 kbps reaches a median SI-SDR of -1.0 dB, the first bitrate at which target-signal energy and reconstruction- error energy are roughly comparable. The gap to Encodec at 1.6 kbps is around 22 dB and at 12.8 kbps around 13 dB, larger in both cases than the ~ 6 dB Encodec itself attributes to adversarial training in their own ablation (Défossez et al., 2022, §3.5). The residual several dB across the curve presumably reflect the other differences in data, scale, and architecture detail. Within our own setup, the gap-narrowing trend is the more interesting object than the absolute size of the gap, because it is internally consistent: the same codec evaluated under the same protocol approaches the with-GAN reference more closely as the bottleneck loosens.

A complementary view of the perceptual-loss negative result links this back to §4. The swap-STFT collapse is a clean falsification: in a setting where the perceptual loss is the only spectral signal available to the model, training fails to reach even baseline-level reconstruction. The additive variant's near-equivalence to baseline therefore cannot be attributed to "perceptual is doing the work that STFT used to do". The perceptual loss is simply weaker than STFT in the envelope dimension and provides nothing in the phase dimension, leaving the codec with the same gradient signal it had before.

The cleanest experimental follow-up is a within-architecture GAN ablation, which would let us decompose the gap-narrowing trend in Fig. 2a into a controlled adversarial contribution and the residual data / scale / architecture confound. The conceptually deeper question, raised by the structural argument in §4, is whether a *phase-aware* self-supervised audio encoder is achievable as a pretraining target. Our HuBERT result identifies the gap negatively; constructing pretext tasks whose representations must encode phase, and evaluating the resulting features as a non-adversarial perceptual loss, is an open problem we believe is worth its own line of work.

7 Limitations and future work

Several limitations qualify the conclusions above. The most significant is that we use no adversarial discriminator, by design, which caps how strong the absolute SI-SDR numbers from this architecture can be and means the bitrate-scan curve sits below any system that does use one. Each bitrate point is also a single seed and a single training run, so we have no within-condition variance estimates against which to gauge the per-octave gain we report. Evaluation is on speech alone; generalisation to music is not characterised. Finally, the Encodec reference numbers in Fig. 2a are read approximately from a published table on a different setup (different sample rate, training corpus, and model size), so the gap-narrowing trend in our comparison can only be motivated, not concluded, until the controlled within-architecture ablation we propose below has been run.

The most direct experimental follow-up is precisely that controlled GAN ablation: training the same architecture with and without an adversarial discriminator at all four bitrates would let us decompose the gap-narrowing trend into a clean adversarial contribution and a residual confound. Beyond that, our HuBERT negative result motivates a longer-horizon question: whether a self-supervised pretraining objective can be designed so that the resulting representations must encode phase information, providing a non-adversarial path to the phase signal that this work has shown ASR-style SSL features cannot supply. Smaller follow-ups also suggest themselves. Quantizer dropout (Zeghidour et al., 2021) would let a single trained model support multiple inference bitrates rather than the four independent models we trained here. Phase-specific differentiable losses such as instantaneous frequency or group-delay distortion are an alternative to adversarial training that does not require a discriminator network. And evaluation on a music-domain dataset would characterise the codec's transfer behaviour beyond clean read speech.

References

A. Défossez, J. Copet, G. Synnaeve, and Y. Adi. High fidelity neural audio compression. *arXiv preprint arXiv:2210.13438*, 2022.

W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, and A. Mohamed. HuBERT: Self-supervised speech representation learning by masked prediction of hidden units. In *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, volume 29, pages 3451–3460, 2021.

R. Kumar, P. Seetharaman, A. Luebs, I. Kumar, and K. Kumar. High-fidelity audio compression with improved RVQGAN. *Advances in Neural Information Processing Systems*, 36, 2023.

T. Salimans and D. P. Kingma. Weight normalization: A simple reparameterization to accelerate training of deep neural networks. In *Advances in Neural Information Processing Systems*, 2016.

A. van den Oord, O. Vinyals, and K. Kavukcuoglu. Neural discrete representation learning. In *Advances in Neural Information Processing Systems*, 2017.

N. Zeghidour, A. Luebs, A. Omran, J. Skoglund, and M. Tagliasacchi. Soundstream: An end-to-end neural audio codec. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 30:495–507, 2021.